

tier_fw_qemu_smoke

tier_fw_qemu_smoke.sh — QEMU aarch64 firmware boot smoke (emulation-fabric P2 / Tfw tier)

Field	Value
Revision	1
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Status	active

Overview

First deliverable of emulation-fabric **P2 (Tfw tier)** per [../design/emulation_fabric/ROADMAP.md](#) (and the Tfw row of [../design/emulation_fabric/DESIGN.md](#)): prove the firmware/Linux-target tier is operational on the macOS dev host by booting a generic **aarch64 -machine virt** target under `qemu-system-aarch64` with **edk2 UEFI** firmware (TCG by default; HVF accel optional) and asserting a **real boot milestone string** captured from the serial console — the edk2/UEFI firmware banner, a U-Boot prompt, or a Linux kernel boot string. This is the boot floor that the containers/pkg/vm U-Boot slot-switch logic builds on (REPORT §3.1).

It is **not** an Android A/B tier: `-machine virt` is a generic ARMv8 board, not an RK3588 SoC model (no Rockchip clocks/PMIC/Mali) — REPORT §3.1 + §8 honest gap.

Prerequisites

- `qemu-system-aarch64` — `brew install qemu`.
- An **edk2 aarch64 UEFI** firmware blob. The `qemu` Homebrew formula ships one at `$(brew --prefix qemu)/share/qemu/edk2-aarch64-code.fd`; the script auto-detects it (and the common `QEMU_EFI.fd` / `AAVMF_CODE.fd` names) under the `qemu` share dirs, or accepts `QEMU_EFI=/path/....`.
- macOS host (REPORT §0): TCG always works; HVF is optional acceleration (`QEMU_ACCEL=hvf`).

Usage

```
bash tests/emulator/tier_fw_qemu_smoke.sh
QEMU_EFI=/opt/homebrew/share/qemu/edk2-aarch64-code.fd bash tests/
    emulator/tier_fw_qemu_smoke.sh
BOOT_TIMEOUT=45 QEMU_ACCEL=hvf bash tests/emulator/
    tier_fw_qemu_smoke.sh
```

Env inputs (all optional): QEMU_BIN, QEMU_EFI, BOOT_TIMEOUT (default 60s), QEMU_MEM (default 512M — bounded per §12.6), QEMU_ACCEL (default tcg).

Anti-bluff contract (§11.4 / §11.4.3)

- **PASS (exit 0)** requires a **real milestone string** captured from the serial log of an actually-launched qemu process — matched case-insensitively against `UEFI Interactive Shell | EDK II | BdsDxe | TianoCore | U-Boot 20 | Hit any key to stop | Booting Linux | Linux version | Synchronous Exception`. Never a metadata-only or exit-code-only pass.
- **SKIP (exit 3)** when a prereq is **genuinely absent** (qemu not installed, or no `UEFI` firmware blob) — prints exactly what is missing + the install command. This is the honest §11.4.3 fallback, NOT a fake pass.
- **FAIL (exit 1)** when qemu launched but no milestone appeared within the timeout (cites the serial + stdout log paths).

Evidence (under `docs/qa/<run-id>-qemu-fw-smoke/`): `qemu_serial.log` (the guest console capture), `qemu_stdout.log` (accel evidence), and `boot_smoke_result.txt` (qemu/efi/accel/booted/milestone + first serial lines).

Edge cases / internal behaviour

- **qemu --prefix is not qemu installed:** `brew --prefix qemu` reports the formula's would-be location even when qemu is NOT installed; the script keys off the actual binary (command `-v / $QEMU_BIN` executable check), not the prefix — §11.4.6 captured as FACT (this host: `qemu` absent → SKIP).
- **No disk image:** the firmware boots and (finding nothing bootable) lands at the `edk2` `UEFI` shell / boot-failure prompt — both emit the firmware banner on the serial console, which is the asserted milestone. `-no-reboot` stops a boot-fail reboot loop.
- **Cleanup (§11.4.14 / §12):** the script captures the qemu pid (\$) and ALWAYS kills **only that pid** on exit (graceful kill, then `kill -9` after a bounded wait) — NEVER a broad `pkill qemu.*`, which could touch podman's qemu or any other qemu instance.
- **Bounded resource use:** 512M guest RAM by default (§12.6 host-memory ceiling); single short-lived process.

Related

- `docs/research/emulation_infra/REPORT.md` (§0 host fact, §3.1 QEMU virt+edk2 verdict, §8 RK3588-SoC gap)

- docs/design/emulation_fabric/{DESIGN,ROADMAP}.md (Tfw row / P2 milestone)
- tests/emulator/tier1_avd_hvf_smoke.sh (sibling P1/T1 AVD-on-HVF smoke)
- containers/pkg/vm (QEMU aarch64 virt+UEFI orchestration the Tfw tier feeds)

Last verified: 2026-06-10 — P2 ACTIVE: qemu 11.0.1 installed, real aarch64 virt+edk2 UEFI boot reached the BdsDxe/UEFI-Shell milestone (out-of-the-box, qemu on PATH). prereq-absent skip, evidence written). Boot/milestone/cleanup path independently exercised via a fake-qemu harness emitting a real edk2 banner: PASS on the captured milestone + the live pid killed with no leak (harness probe-dir not committed).